

MATH0111 Frontiers in Computational Mathematics

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0111
<i>Level:</i>	7 (UG), 7 (PG)
<i>Normal student group(s):</i>	UG: Y4 Mathematics programmes PG: MSc Mathematics
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	85% examination, 15% coursework
<i>Normal Pre-requisites:</i>	MATH0033 (Numerical Methods) or MATH0058 (Computational Methods)
<i>Lecturer:</i>	Dr M Jensen, Prof E Burman, Prof T Betcke

Course Description and Objectives

This module will introduce state-of-the-art numerical methods for solving complex problems in mathematics, science, and technology. Topics range from advances in the theory of numerical analysis to mathematical challenges arising in high-performance computing. The course aims to deepen understanding of numerical techniques in current research and real-world applications through topical academic texts and research papers. The module is split into three blocks, each introducing a separate, current development in Computational Mathematics. Typically, each block is taught by a different lecturer to take full advantage of the expertise in Computational Mathematics at UCL.

Recommended Texts

- A Ern and JL Guermond, *Finite Elements I: Approximation and Interpolation* <https://link.springer.com/book/10.1007/978-3-030-56341-7>
- A Ern and JL Guermond, *Finite Elements II: Galerkin Approximation, Elliptic and Mixed PDEs* <https://link.springer.com/book/10.1007/978-3-030-56923-5>
- A Ern and JL Guermond, *Finite Elements III: First-Order and Time-Dependent PDEs* <https://link.springer.com/book/10.1007/978-3-030-57348-5>
- M Neilan, AJ Salgado, and W Zhang, *Numerical analysis of strongly nonlinear PDEs* <https://doi.org/10.1017/S0962492917000071>
- PG Martinsson and JA Tropp, *Randomized numerical linear algebra: Foundations and algorithms* <https://doi.org/10.1017/S0962492920000021>

Detailed Syllabus

The focus of the topics is subject to change from year to year, with the inclusion of guest lecturers and new research developments, but examples may include:

1. Numerical Analysis
2. Numerical Linear Algebra
3. Numerical Solutions of Differential Equations

4. Optimisation and Nonlinear Equations
5. Data-driven Scientific Computing
6. Computational Mathematics for Real-World Problems

September 2026